PREVENTATIVE MAINTENANCE CHECKLIST

Enclosed Cab	Underground Haulage Truck	Tier 4	Dump

1000	Hour

DATE: _____

UNIT NUMBER: _____

TECHNICIANS: _____

HOURS: _____

WORK ORDER: _____

***** PM TO BE COMPLETED ONCE THE UNIT IS WASHED *****

Note: All specifications, procedures, and parts not identified in this PM can be found in the OEM manuals

PRE-WORK SAFETY TASKS

MANDATORY PPE

LOCKOUT / TAGOUT – Ensure all procedures are followed to eliminate stored energy

Before working on brake system or hydraulic system drain accumulator pressure

Ensure deck, walkways and steps are free from foreign material such as oil and debris and that traction surface is in place and in good condition

Read and understand all warning plates on machine before operating, lubricating or repairing machine

To avoid burns, be alert for hot parts on machines, which have just been stopped and have hot fluids in lines, tubes and compartments.

Battery electrolyte contains acid – avoid contact with skin or eyes.

Perform all running checks in an open area free of obstruction and personnel

Ensure the box is locked in the raised position and the steering lock is in place before working on the machine

Ensure the box is free of muck before working on the machine

If working on air system brakes, bleed down air system

Ensure moldboard and ripper are lowered to the ground before servicing equipment

Refer to manufacturer's operation and maintenance manuals for equipment specifications, fluid capacities, operating procedures and maintenance procedures

Be sure the area around the machine is clear of personnel and obstructions while performing running checks

FOR MORE COMPLETE LIST OF SAFETY MESSAGES, REFER TO THE OPERATION AND MAINTENANCE MANUAL FOR THIS MACHINE

0 = OK, N = DEFECT FOUND AND REPAIRED, C = DEFECT FOUND NOT REPAIRED, R = N/A

		0	N	C	R
	⚠ ALL TASKS MUST BE ACCOMPLISHED ⚠ ONLY BY QUALIFIED PERSONNEL				
	PRE-OPERATOR TASKS AND INITIAL START-ASSESSMENT				
	Mark axle (remove bolt) gap before making engine setup				
	Daily book - check who has book & ensure book is current & remote				
	Daily book comments - review				
	Check condition of guard walls, ensure they all unfold and lock in				
	Check "OTHER ENERGY" labels, make sure they are in place and legible				
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	Check cab door - ensure door opens and closes easily, that hinge is good				
	Check cab door - ensure door opens and closes easily, that hinge is good				
	Check condition & operation access steps, lids & prop rods, satisfy				
	Wipe all exterior surfaces are in good shape and the steps and grab				
	Check for any fluid leaks				
	Check frame & chassis for cracks & report damage				
	Check the frame rub bars for excess wear				
	OPERATING PACKING				
	Check proper operation of the back state beacon, calibrate tank and backup alarm				
	Test the operation of the reverse/ forward (fwy) to start machine while in neutral				
	Test the operation of the function lockout				
	Gauges, dash lights, marker lights, steering lights - check condition & operation				
	Operator control labels - ensure they are in place & legible				
	Check the condition of the operator seat (are spring assembly and ensure the seated stops are in good condition)				
	Labels - inspect labelling on unit (CIM, marking, operator labels)				
	Brake test pendants please - check if in in place				
	Brake pedal - check for smooth operation, wear of pivot & remove any obstructions				
	Ensure ABS is functioning properly				
	Service brake holding capacity - test				
	Park brake holding capacity - test				
	Tonnetta - check for smooth operation & remove any obstructions on				
	Accelerator - check for smooth operation & remove any obstructions on floor				
	Surface RIM download, provide download file to Flamenco / Reliability				
	Seat and armrests - check overall condition and ensure the height and				
	Check condition of the seat belt				
	Seatbelt - check condition & expiry date				
	ABS / ESC - check for damage & loose mounting bolts				
	Hydraulic & transmission control levers - check for free movement & looseness				
	Check condition of the door, hinges, latches, locking setup, and door seals				
	All leaks in cab & under door panel - check for leaks				
	Check condition of ambulance stepes and verify the steps and markers are working				
	Fire extinguisher check condition and ensure the emergency egress hab is in place				
	Check operation of the heating / air conditioning system including the fresh air intake, ensure the air intake is free from obstructions, ensure the cleanliness of the cab and ensure the air pressure and in good condition				
	ENGINE-CHECKS				
	Engine breather tube - check for excessive smoke or leaking engine oil				
	Transmission - check for oil leaks				
	Exhaust / Fission - operate unit, listen for noise and ensure that all				
	Perform exhaust system				
	Perform engine breather safe and call test				
	Verify operation of the Kempa lockup				
	Verify all hydraulic functions are working, including flow				
	Check operation of the payload monitoring system, calibrate system				
	Ensure the box up gear lockout is working correctly.				
	Verify operation of the heavy exhaustor / compression brake				
	Check for play in the articulation pins, ensure that all of the bolts are nuggs				
	Steering test - check clearance both ways to avoid steering cylinders hammering out, adjust steering stops and neutralise valve if				
	Check steering cycle times at high idle (sec)	left to begin	right to left		
	Verify secondary (emergency) steering is operational				
	Check the articulation pins, ensure that all of the steering cylinder pins and bushings. Ensure pin retainers are tight and in place.				
	Check for play (including side to side play) in the left cylinder pins and bushings. Ensure pin retainers are tight and in place.				
	Check lift cylinder cycle times (sec)	raised to fully raised	fully raised to raised		
	Check tilt cylinder drift rate				
	Check the operation of the backup camera, ensure the camera is aimed correctly. Reference pylon A under camera, pylon C at 6' (2m) directly in line with C, and pylon D directly in line with A and C at far end, so that the Positive pylons A and D at 6' (2m)to the right and left of the camera centerline respectively and halfway between pylons A and D. Negative pylons B and C at 6' (2m) in line left and right respectively with the camera. Note: Position as shown in illustration. Camera view (illustration 2) shows the view from the camera. Pylon A must be in the middle of the camera view, Pylon A and B must be the same distance to the outside of the monitor as per arrows in camera view, Pylons A and B must not be seen by the camera.				
	Surface forced regeneration				
	Check brake accumulator charge valve settings.				
	Surface pilot all accumulator leak down test				
	Surface pilot all accumulator pressure test				
	Check steering relief pressure				
	Check dump relief pressure				
	Turn park brake the release system.				
	ISOLATION				
	Lift machine and place on jack stands (or ramps)				
	Bleed hydraulic and brake pressure.				
	Install steading lock and hook lock				
	Tag and lockout machine				
	Barricade working area.				
	Ensure cable tilt function is working properly and that the cylinder				
	Install safety lock				
	Install safety tilt lock				
	OIL SAMPLING + OIL AND FILTER CHANGES				
	Take engine oil sample, change oil, inspect magnetic plug.				
	Check magnetic plug from 1 - 5 based on "Magnetic Plug Ratings"				
	Replace engine oil filter, run old filter and inspect				
	Check oil debris from 1 - 5 based on "Filter Contamination Rating"				
	Replace water separator & fuel filter				
	Replace engine fuel filter				
	Engine coolant - take sample, check coolant level				
	Replace engine coolant filter				
	Change Transmission filter, run old filter and inspect				
	Check oil debris from 1 - 5 based on "Filter Contamination Rating"				
	Clean transmission magnetic screen				
	Take sample and change transmission oil, inspect magnetic plug				
	Check magnetic plug from 1 - 5 based on "Magnetic Plug Ratings"				
	Change gearbox filter, run old filter and inspect				
	Check oil debris from 1 - 5 based on "Filter Contamination Rating"				
	Take sample and change gearbox oil, inspect magnetic plug				
	Check magnetic plug from 1 - 5 based on "Magnetic Plug Ratings"				
	Replace Crankcase/ Fumes disposal filter (CVD)				
	Clean ABS filter screen				
	Change hydraulic (return) filter, run old filter and inspect				
	Check oil debris from 1 - 5 based on "Filter Contamination Rating"				
	Change brake filter, run old filter and inspect				
	Check filter debris from 1 - 5 based on "Filter Contamination Rating"				
	Take sample and change hydraulic oil, inspect magnetic plug				
	Check magnetic plug from 1 - 5 based on "Magnetic Plug Ratings"				
	Check and clear DIFFERential oil - take sample, change oil, check				
	Check magnetic plug from 1 - 5 based on "Magnetic Plug Ratings"				
	Take oil sample, change final drive oil, inspect magnetic plug				
	Check magnetic plug from 1 - 5 based on "Magnetic Plug Ratings"				
	Take sample and change final drive oil (all four corners), inspect magnetic plugs				
	Ensure proper labelling and packaging of samples to be sent out for				
	Replace primary & secondary air filters.				
	Replace cab air filter				
	Replace recirculating air (A/C) filter				
	ENGINE COMPARTMENT				
	CONTACTOR - check condition & tightness of mount bolts, inspect fins for damage and verify the fins are not plugged and the cooks are not loosening. Ensure the contactor is secure and not loosening.				
	AC damage and verify the fins are not plugged and the cooks are not loosening.				
	HYDRAULIC COOLER - check condition & tightness of mount bolts, inspect fins for damage and verify the fins are not plugged and the cooks are not loosening.				
	AUX OIL COOLER - check condition & tightness of mount bolts, inspect fins for damage and verify the fins are not plugged and the cooks are not loosening.				
	AIR CONDITIONING CONDENSER - check condition & tightness of mount bolts, inspect fins for damage and verify the fins are not plugged and the cooks are not loosening, and ensure fans are secured to the condenser body and mounts.				
	ENGINE - check condition of overall system, including filter housing and pre-cleaner, piping to and from engine and aftertreatter, and the intake manifold (check all components for loose clamps, bolts, nuts, and ensure that all components are secure and in place)				
	EXHAUST - check condition of overall system including muffler, clamps, gaskets and exhaust drain on piping				
	Check the intake and exhaust connections and check condition of hoses, bolts (bearings, fasteners, and bushings), and ensure there is				
	Check torque on impeller and bushing housing clamps (loose clamps may be a quick indicator of internal issues) and check for any evidence of				
	BATTERIES - check all terminals and posts are free from corrosion and securely fastened				
	Check the electrolytes in the battery and test specific gravity (if battery is not maintenance free)				
	Check for loose or corroded wiring on battery and master disconnect				
	Check condition of all sinking lose condition & studing				
	Check starter mounting bolts and wiring connections.				
	Check for loose or corroded wiring on starter				
	Check alternator mounting bolts and wiring.				
	Check alternator connections and wiring for alternator				
	Check belt tension and condition				
	Check condition of pulleys, fan and shrouds				
	Check condition of all hoses for leaks.				
	Check condition of engine mounts & tightness of bolts.				
	Inspect fuel lines for leaks and clearance, Inspect fitment of all clamps				
	Check for leaks from water pump, hoses, thermostat and coanol				
	Check condition of all hoses				
	Check condition of engine covers and check all guards are in place & secure				
	All & fuel leaks - check engine bay for leaks & cleanliness.				
	Hydraulic, transmission and brake/steering pumps - check for leaks				
	TIRES				
	Check tires for large sections missing from the face, face peeling, large cuts (cuts exposed), sidewall damage (impact damage), sidewall cuts (cords exposed), or any tire bulging				
	Check condition of rims for cracks, bends, missing or loose studs /				
	Check condition of wheel nuts, ensure they are tight				
	Please record outer sidewall Year Tire Stamp				
	Measure inside & outside tire depth, record lowest reading (in 16/4")				
	Check tire pressure to ensure				
	POWERTRAIN				
	Check condition of transmission mounts & torque securing bolts				
	Check for play in the articulation bearing, ensure that all of the bolts are tight				
	Inspect all universal joints for dust and play, check & tighten bolts.				
	Check tightness of oscillating axle bolts, check for play in the				
	Check front, center, and rear axle mounting flange pivot pins and bushings for play, ensure they are tapping grease				
	Check front center, and rear, stabilizer links for excess play or				
	Check suspension cylinders for leaks or damage, measure suspension				
	Check front and rear axle mounting bolts using torque (tapping) to identify any loose bolts.				
	Check torque / Reconnection compartment for any leaks, check hoses for				
	Check condition of steering, check directional harmonics for proper steering and any abnormal				
	Check condition of ground stops in the articulation joints				
	DUMP BOX / LOADING BOX				
	Check box for dent and cracks				
	Check for uneven wear or gaps on box wear support pads.				
	Ensure grease fitting are in good condition and all pins are tapping				
	Inspect condition dump cylinders				
	Take and record 5 measurements of from the position indicated (Ensure the box is empty and the truck is level on				
	Pos 1 Left Front Fender Top	Actual			
	Pos 2 Front Floor	Actual			
	Pos 3 Right Front Fender Top	Actual			
	Pos 4 Left Center Fender Top	Actual			
	Pos 5 Center Floor	Actual			
	Pos 6 Right Center Fender Top	Actual			
	Pos 7 Left Rear Fender Top	Actual			
	Pos 8 Rear Floor	Actual			
	Pos 9 Right Rear Fender Top	Actual			
	HYDRAULIC AND HYDRAULIC TANK COMPARTMENT				
	Check condition & cracking of all hoses in hydraulic tank bay				
	Check condition & cracking of all hoses in articulation area.				
	Check condition & cracking of all hoses in rear frame area				
	Check condition & installation of hydraulic valves				
	Check hydraulic tank condition & mounting, check for leaks				
	Check all valves for leaks				
	GREASING				
	Check level of the autogreaser reservoir				
	Check for proper operation of grease pump				
	Check for leaks at all lines & connections on auto lube system, if				
	Ensure fan bearing on engine				
	Check for proper grease distribution at all auto lube points				
	Grease all universal joints, drag links & pitman drive				
	Grease all cover and wear hinges.				
	PART WASH TASKS				
	Perform all fluid levels and verify all drain and level plugs are				
	Take note (say) filters, service parts, hose, and bolt properly closed and locked				
	Check note (say) of debris, old parts, excess / unused materials and				
	Ensure all tools, equipment, and parts, ensure checked for damaged cabinet, etc				
	Return equipment marked as or inspected to a safe operating condition				
	Ensure work orders are submitted to planning for all fluid corrective				
	Ensure all tools are cleaned and stored				
	Ensure all repairs and equipment is signed off, dated and returned to Supervisor / Planning				
	Walk around equipment to ensure that all foreign materials are cleaned up & put away properly. (Rags, tools, drums, chains, parts, garbage)				
	Ensure all equipment is signed off, dated and returned to Supervisor				
	Check work scope sheets are filled out & handed into supervisor for				
	Complete parts & core returns				
	Clean up old filters / oils and dispose of properly				
	RE-ISOLATION				
	Remove lock pin and lower cable, ensure it is secured				
	Remove padlock & danger tags & re-energise equipment				
	Remove articulation lock, tie up lock, lower equipment to the ground and remove barricades				

Y = OK, N = DEFECT FOUND AND REPAIRED, C = DEFECT FOUND NOT REPAIRED, N = N/A	Y	N	C	N/A
⚠ = Tasks required not at obvious ⚠ = CORRECTIVE MONITORING POINT				
START UP				
001 Check work and inspect for any fluid leaks				
002 Verify all fluids levels				
003 Verify operations meet sets in logbook				
SIGN OFF				

TECHNICIAN NAME:

TECHNICIAN SIGNATURE:

SUPERVISOR NAME:

SUPERVISOR SIGNATURE:

DATE:

DEFFICIENCY LIST
CATERPILLAR SCOOP (LHD)
HR - PM

Step No.	Defficiency	Corrective Action Required	Work Request No.	Comp

PM FEEDBACK FORM

HAVE YOU FOUND ANY PARTS DISCREPANCIES?			
STOCK #	***INCORRECT PART #***	***CORRECT PART #***	***DESCRIPTION OF PART***

HAVE YOU IDENTIFIED ANY MAINTENANCE TASK TO BE UPDATED, ADDITION OR DELETED?	
TASK DESCRIPTION:	***UPDATE / ADD / DELETE (circle one)***